28-29-30//NOVEMBRE 2017

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La più importante **conferenza italiana** sulle tecnologie **Microsoft**

WPC 2017 MILANO

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DEV023 - What's new in SQL Server 2017

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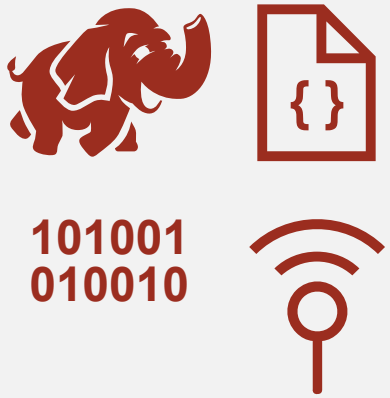
Agenda

- Introduction
- Power BI Report Server
- Machine Learning Services
- SSIS
- Database Engine
- SQL on Linux
- Graph Database
- High Availability & Disaster Recovery

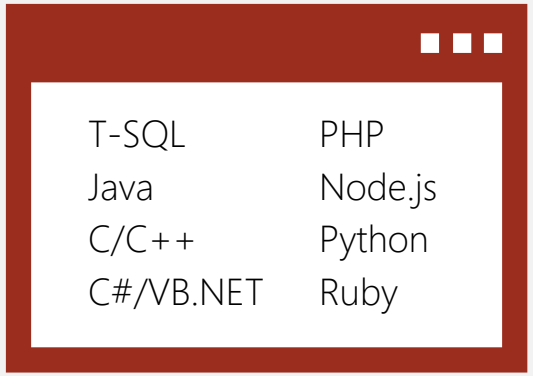
SQL Server 2017

Meeting you where you are

It’s the same SQL Server Database Engine that has many features and services available for all your applications—regardless of your operational ecosystem.



Any data



Any application

Anywhere

Windows Server

Linux

docker

Choice of platform

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SQL SERVER 2017

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INDUSTRY-LEADING PERFORMANCE AND SECURITY NOW ON LINUX AND DOCKER

Choice of platform and language

T-SQL	PHP
Java	Node.js
C/C++	Python
C#/VB.NET	Ruby

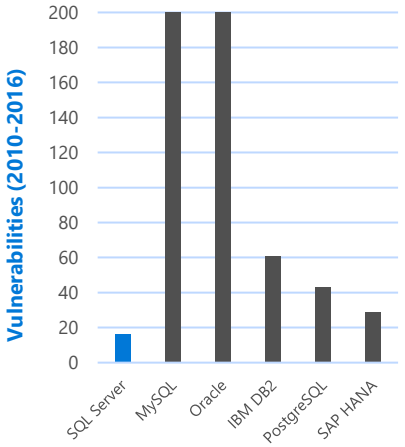
Industry-leading performance

#1 TPC-H performance
1TB, 10TB, 30TB

#1 TPC-E performance

#1 price/performance

Most secure over the last 7 years

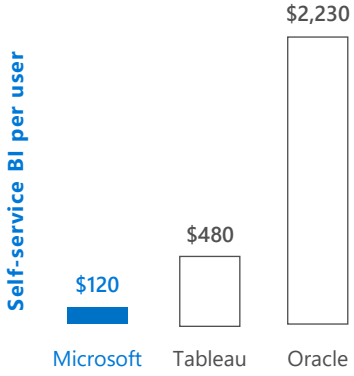


Only commercial DB with AI built-in

R and Python +
in-memory at massive scale

Native T-SQL scoring

End-to-end mobile BI on any device



A fraction of the cost

In-memory across all workloads

Private cloud

Public cloud

Most consistent data platform

SQL SERVER 2017 EDITIONS

Common programming surface area - develop once and scale across editions

Express

SMALL-SCALE APPLICATIONS

- ✓ Development and management tools
- ✓ Easy backup and restore to Microsoft Azure
- ✓ Free to use

Standard

FULLY FEATURED DATABASE FOR MID-TIER APPLICATIONS AND DATA MARTS

- ✓ End-to-end database security with Always Encrypted
- ✓ Enhanced in-memory performance for all workloads
- ✓ Basic reporting
- ✓ Basic analytics
- ✓ Hybrid scenarios: Stretch Database, backup

Enterprise

MISSION CRITICAL IN-MEMORY PERFORMANCE AND SCALE, SECURITY AND HIGH AVAILABILITY

- ✓ Mission critical high availability on Windows and Linux
- ✓ Enhanced in-memory performance
- ✓ Faster performance with Adaptive Query Processing
- ✓ Unparalleled data security
- ✓ PB scale data warehousing
- ✓ End-to-end mobile BI with rich visualizations on all major platforms
- ✓ In-database advanced analytics built-in at scale with R and Python
- ✓ Enhanced hybrid scenarios including Stretch Database, HA, DR and backup
- ✓ Software Assurance benefits include unlimited virtualization, Machine Learning Server for Hadoop, and Power BI Report Server

Developer

DEVELOPMENT AND TESTING

- ✓ Build, test, and demo apps in non-production environments
- ✓ Free to use
- ✓ All Enterprise Edition features available

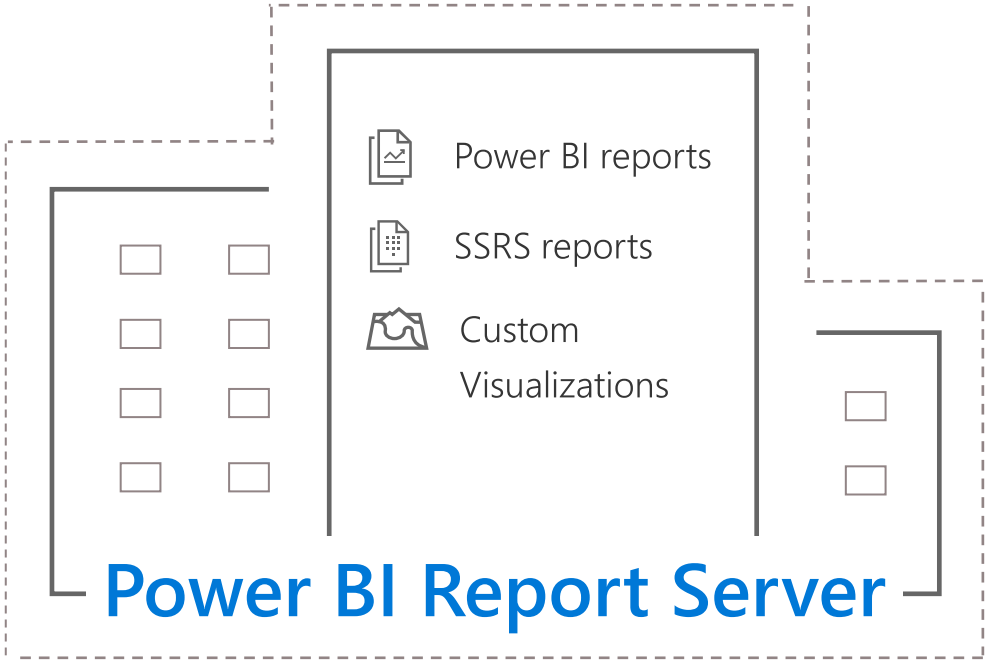
SQL Server 2017 features by edition			Standard	Enterprise
OLTP Performance	Maximum number of cores		24 cores	Unlimited
	Maximum memory utilized per instance		128 GB	OS Max
	Maximum size		524 PB	524 PB
	Advanced OLTP (In-memory OLTP*, Operational analytics)		●	●
	Basic high availability (2-node single database failover, non-readable secondary)		●	●
	Manageability (Management Studio, Policy-Based Management)		●	●
	Enterprise data management (Master Data Services, Data Quality Services)			●
	Advanced HA (Always On - multi-node, multi-db failover, readable secondaries)			●
	Basic Adaptive Query Processing (Interleaved execution)	**NEW***	●	●
	Advanced Adaptive Query Processing (Batch mode memory grant feedback, Batch mode adaptive joins), Automatic Plan Correction	**NEW***		●
Security	Basic security (Always Encrypted, Row-level security, data masking, basic auditing, separation of duties)		●	●
	Advanced security (Transparent Data Encryption)			●
Data Warehousing	Advanced data integration (Fuzzy grouping and look ups, change data capture)			●
	Data warehousing (In-Memory ColumnStore, Partitioning)		●	●
	T-SQL query across relational and Hadoop data with PolyBase**		●	●
Business Intelligence	Programmability & Developer Tools (T-SQL, CLR, Data Types, FileTable, JSON, graph data support)	**NEW***	●	●
	Basic data integration (SSIS, built-in connectors)		●	●
	Basic reporting & analytics*		●	●
	Basic Corporate Business Intelligence (Multi-dimensional models, Basic tabular model, enhanced connectors, new transformations, object-level security, ragged hierarchies)*	**NEW***	●	●
	Mobile BI*			●
	Advanced Corporate Business Intelligence (Advanced tabular model, Direct query, in-memory analytics, advanced data mining)*			●
Advanced Analytics	Basic R and Python integration (Connectivity to R Open and Python, Limited parallelism)*	**NEW***	●	●
	Advanced R and Python integration (Ability to run on GPUs and full parallelism through Machine Learning Services)*	**NEW***		●
Hybrid Cloud	Stretch Database*		●	●

Features in blue are newly available.
*In-memory OLTP subject to memory limits in Standard Edition **Available for Windows Server only

Power BI Report Server

Power BI Report Server

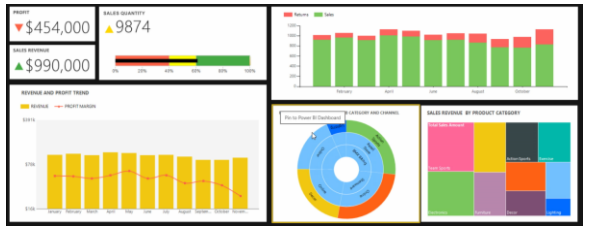
Self-service BI and enterprise reporting in one solution



- Keep Power BI reports on-premises
- Includes all SSRS functionality
- Move to the cloud on your terms

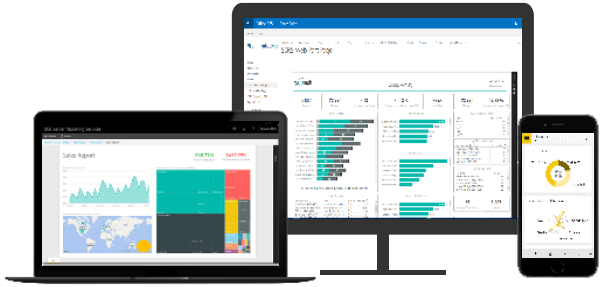
Includes all SSRS functionality

Built on proven SQL Server Reporting Services technology



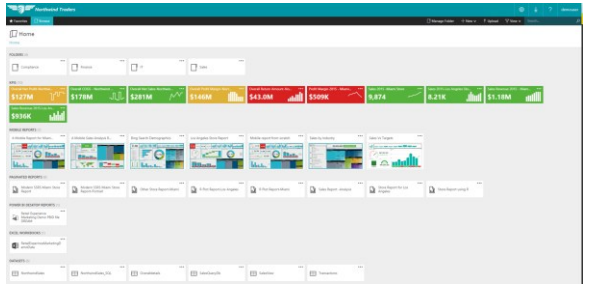
Publish SSRS reports

Create precisely-formatted paginated reports



Consume in multiple ways

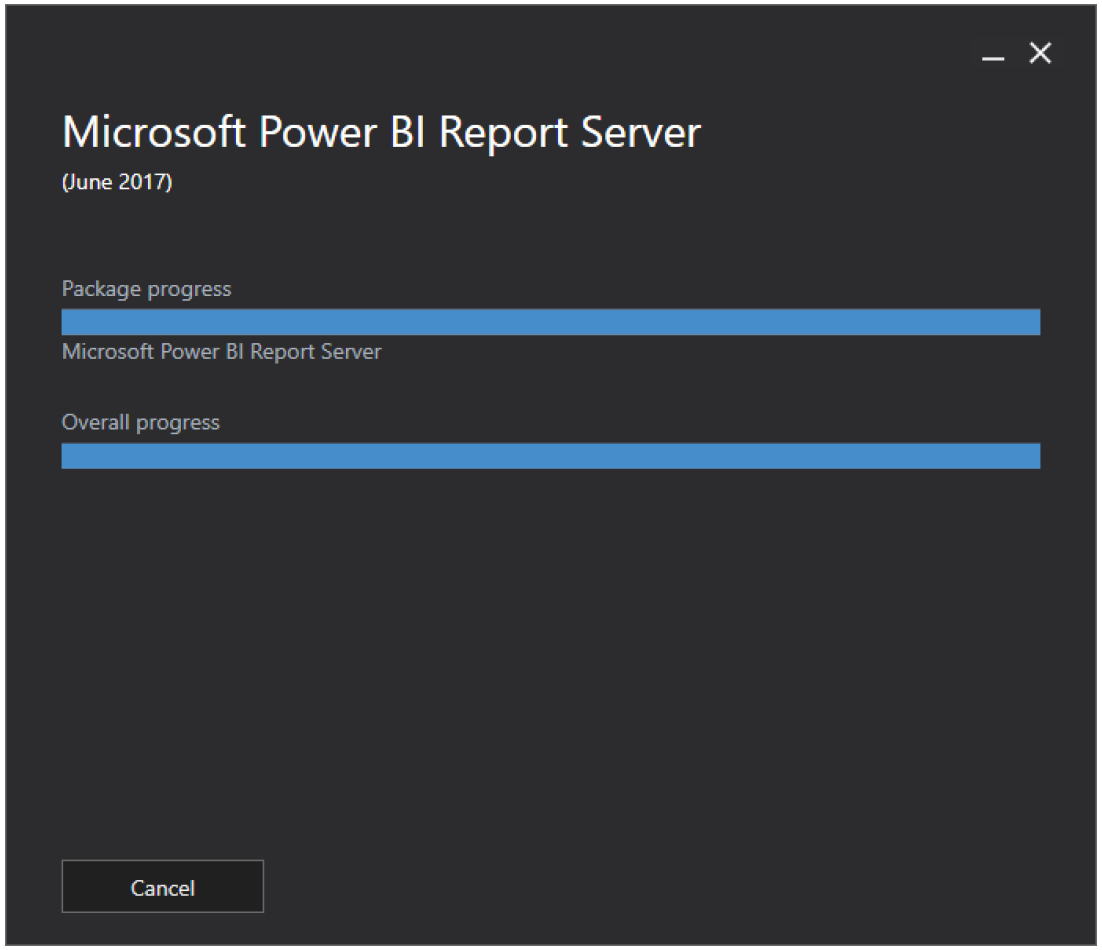
Web browser, mobile app, or embedded into your app



Benefit from an enterprise-ready solution

Organize reports, manage permissions, audit usage

Install in minutes



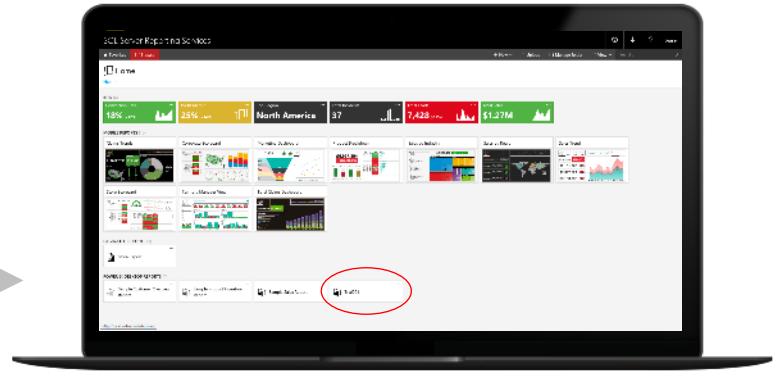
Lightweight download and installation

Zero impact on SQL Server

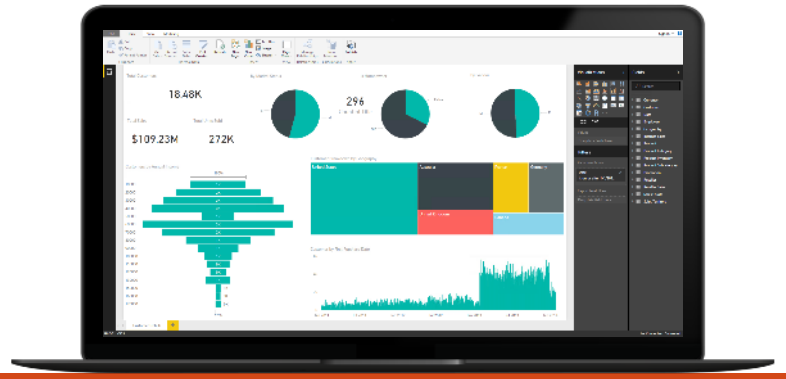
Requirements

- Windows Server 2012+
- SQL Server Database Engine (2008+)
- SQL Server Analysis Services (2012+)

Publish and manage reports



Power BI Report Server



Power BI Desktop

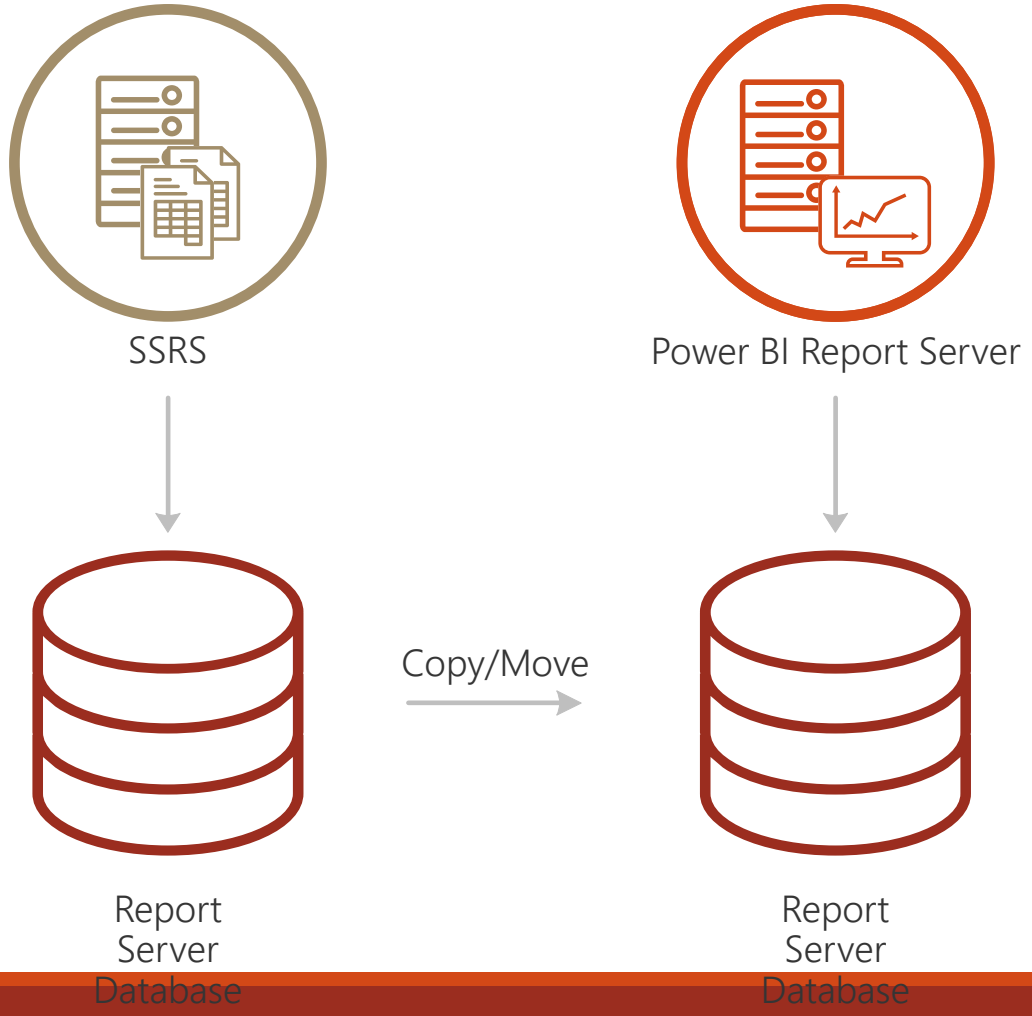
Publish to Power BI Report Server

Organize and manage access to reports

Re-open, edit, and save back

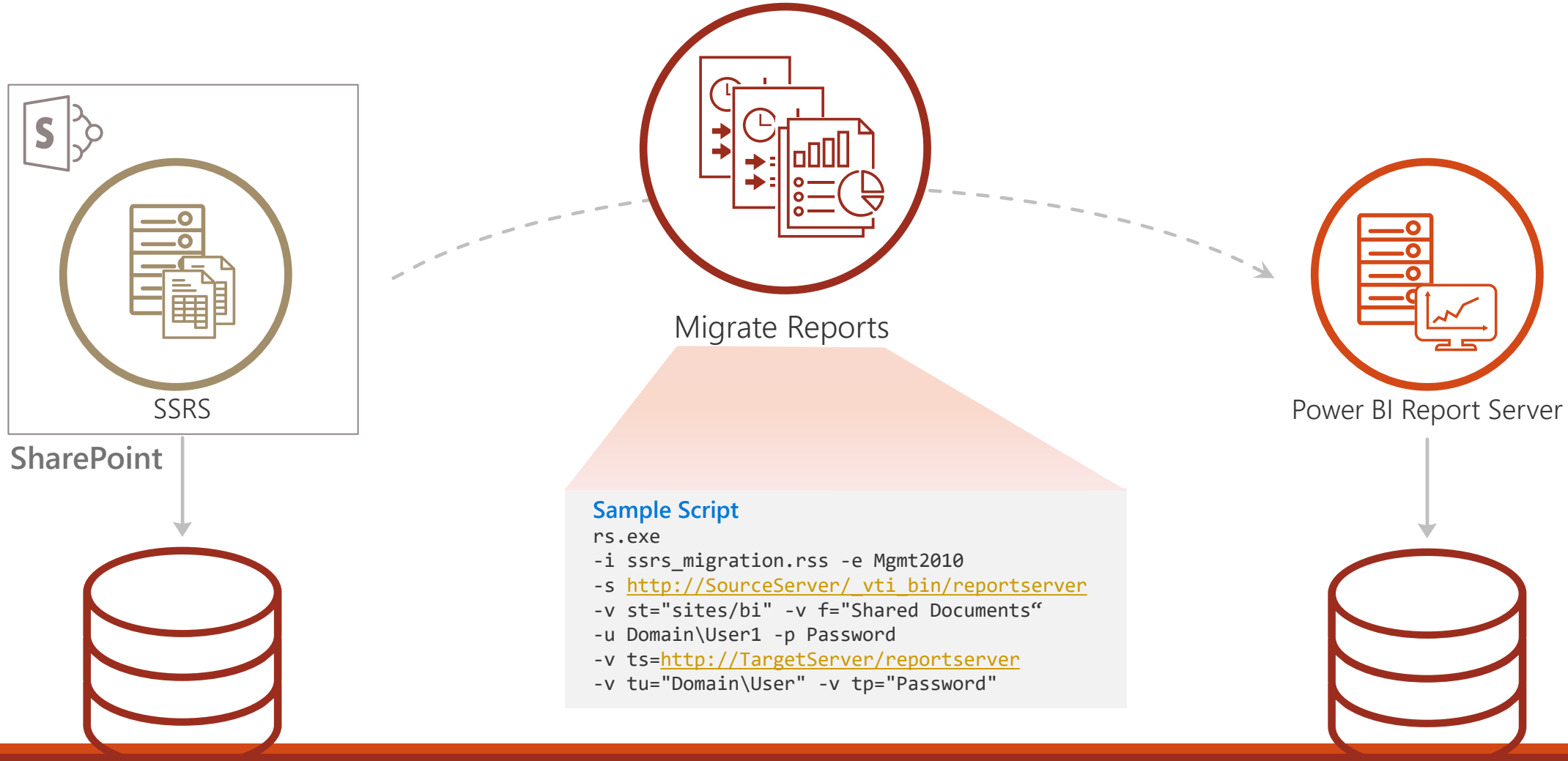
Migration from previous versions

Migrate from SSRS (Native mode)



1. Back up report server database
2. Install Power BI Report Server
3. Connect report server to database using Report Server Configuration Manager

Migrate from SSRS (SharePoint-integrated mode)



Sample Script

```
rs.exe
-i ssrs_migration.rss -e Mgmt2010
-s http://SourceServer/_vti_bin/reportserver
-v st="sites/bi" -v f="Shared Documents“
-u Domain\User1 -p Password
-v ts=http://TargetServer/reportserver
-v tu="Domain\User" -v tp="Password"
```

DEMO

POWER BI REPORT SERVER

Please note that…

- SSRS 2017 is also available as a separate download
- Reporting Services follows the SQL Server licensing model (e.g. if I have SQL STD I have the right to use SSRS STD features)
- Power BI Report Server requires Power BI Premium license and follows that licensing model. In addition, Power BI Pro is required for authoring
  - https://powerbi.microsoft.com/en-us/calculator/

In essence, Power BI Report Server is SSRS + local Power BI + sharing with mobile devices

SQL Server Machine Learning Services

In-Database analytics with SQL Server

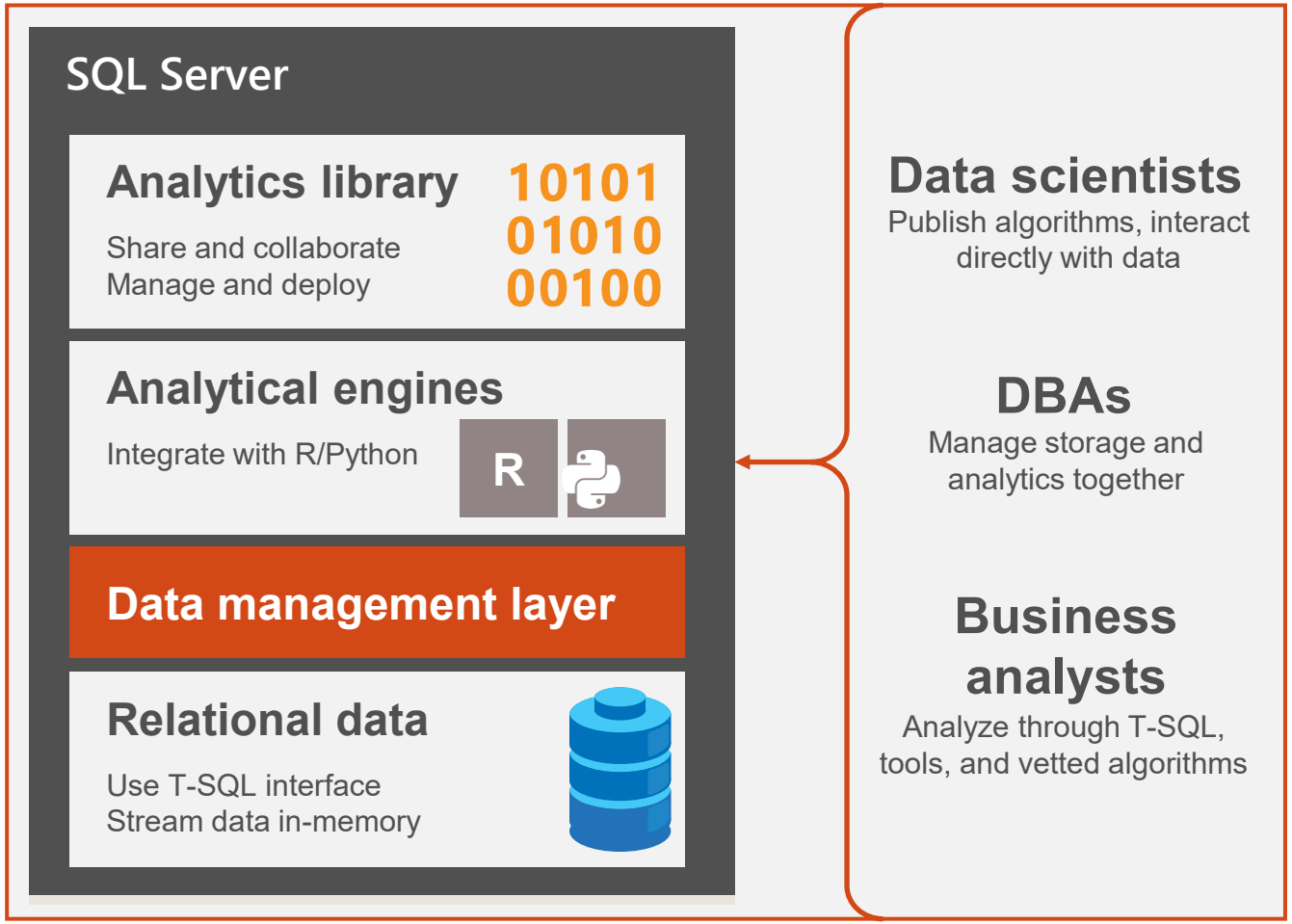
In SQL Server 2016, Microsoft launched two server platforms for integrating the popular open source R language with business applications:

- SQL Server R Services (In-Database), for integration with SQL Server
- Microsoft R Server, for enterprise-level R deployments on Windows and Linux servers

In SQL Server 2017, the name has been changed to reflect support for the popular Python language:

- SQL Server Machine Learning Services (In-Database) supports both R and Python for in-database analytics
- Microsoft Machine Learning Server supports R and Python deployments on Windows servers—expansion to other supported platforms is planned for late 2017

Machine Learning Services



Capability

- Extensible in-database analytics, integrated with R/Py, exposed through T-SQL
- Centralized enterprise library for analytic models

Benefits

- No data movement, resulting in faster time to insights
- Real-time analytics on transactional data
- Integration with existing workflows
- Unified governance across analytics and storage

Enhanced Machine Learning Services (SQL Server 2017)

- Python support
- Microsoft Machine Learning package included
  - https://docs.microsoft.com/en-us/machine-learning-server/r/concept-what-is-the-microsoftml-package
- Process multiple related models in parallel with the **rxExecBy** (in RevoScaleR) function
- Package management at instance level, manage permissions to use packages at database level
- Native scoring with T-SQL **PREDICT**
- In-place upgrade of R components

Setup and configuration

SQL Server 2017 setup

- Install Machine Learning Services (In-Database)
- Consent to install Microsoft R Open/Python
- Optional: Install R packages on SQL Server 2017 machine

Database configuration

- Enable R language extension in database
- Configure path for RRO runtime in database
- Grant EXECUTE EXTERNAL SCRIPT permission to users

```
CREATE EXTERNAL EXTENSION [R]
    USING SYSTEM LAUNCHER
    WITH (RUNTIME_PATH =
    'c:\revolution\bin')
```

```
GRANT EXECUTE SCRIPT ON
EXTERNAL EXTENSION::R TO
DataScientistsRole;
/* User-defined role / users */
```

External script usage from SQL Server

Original R script:

```
IrisPredict <- function(data, model){
 library(e1071)
 predicted_species <- predict(model, data)
 return(predicted_species)
}


library(RODBC)
conn <- odbcConnect("MySqlAzure", uid = myUser, pwd = myPassword);
Iris_data <-sqlFetch(conn, "Iris_Data");
Iris_model <-sqlQuery(conn, "select model from my_iris_model");
IrisPredict (Iris_data, model);
```

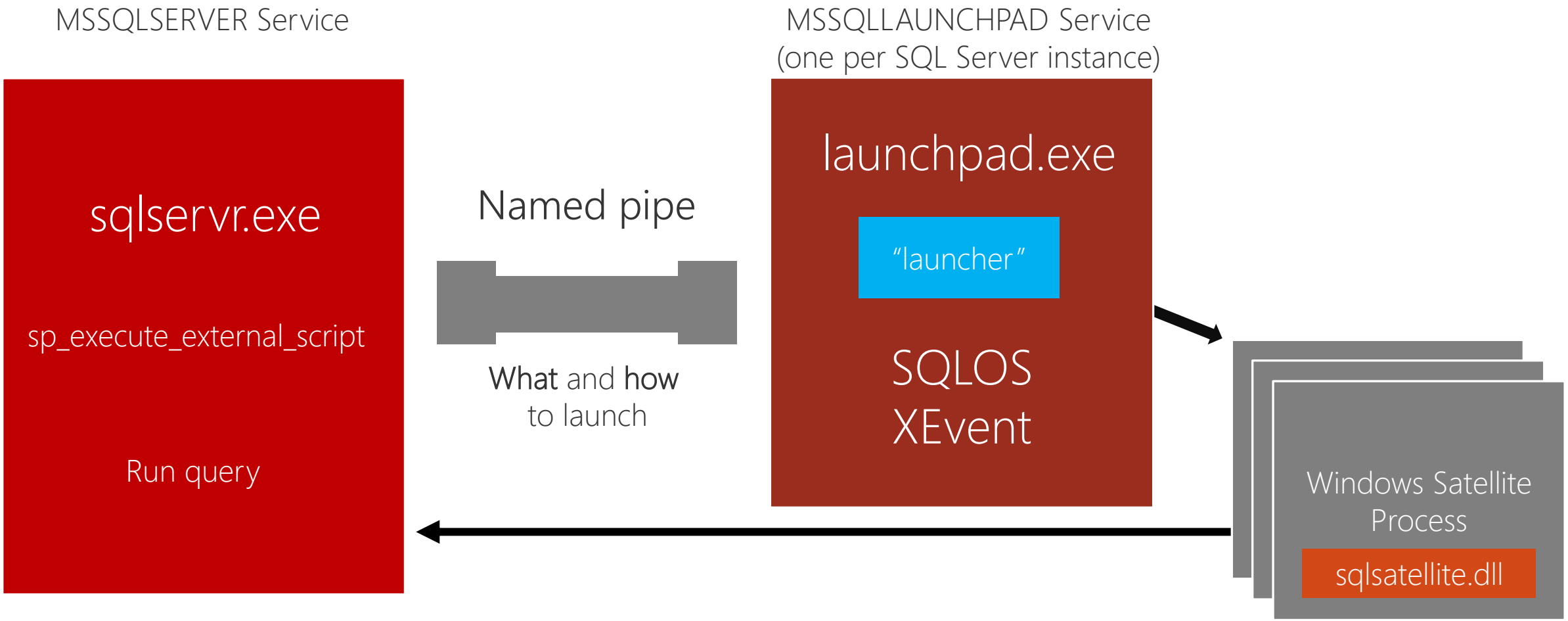
Calling R script from SQL Server:

```
/* Input table schema */
create table Iris_Data (name varchar(100), length int, width int);
/* Model table schema */
create table my_iris_model (model varbinary(max));

declare @iris_model varbinary(max) = (select model from my_iris_model);
exec sp_execute_external_script
  @language = 'R'
, @script = '
IrisPredict <- function(data, model){
 library(e1071)
 predicted_species <- predict(model, data)
 return(predicted_species)
}
IrisPredict(input_data_1, model);
'
, @parallel = default
, @input_data_1 = N'select * from Iris_Data'
, @params = N'@model varbinary(max)'
, @model = @iris_model
with result sets ((name varchar(100), length int, width int
, species varchar(30)));
```

- Values highlighted in yellow are SQL queries embedded in the original R script
- Values highlighted in aqua are R variables that bind to SQL variables by name

The SQL extensibility architecture



SQL Server Machine Learning Services is secure

Reduced surface area and isolation

"external scripts enabled" is required

Script execution outside of SQL Server process space

Script execution requires explicit permission

sp_execute_external_script requires EXECUTE ANY EXTERNAL SCRIPT for non-admins

SQL Server login/user required and db/table access

Satellite processes have limited privileges

Satellite processes run under low privileged, local user accounts in the SQLRUserGroup

Each execution is isolated — different users with different accounts

Windows firewall rules block outbound traffic

Realtime Predictions using Native Scoring

- PREDICT function
  - No dependency on R or Python runtime
  - Enabled on SQL Server in Windows & Linux
- Usage
  - Single or small number of rows scoring
  - Highly concurrent scoring scenarios
  - Predict during INSERT, UPDATE, MERGE statements
- Requirements
  - Models built using RevoScaleR or revoscalepy
    - rxLinMod, rxLogit, rxBTrees, rxDTree, rxDForest
  - Serialized using rxSerializeModel (R) or rx_serialize_model (Python)

PREDICT syntax

```
PREDICT ( MODEL = @model | model_literal,
          DATA = object AS <table_alias> )
   WITH ({ {column_definition } [,...n ] } )

INSERT INTO loan_applications
    (c1, c2, c3, c4, risk_score)
SELECT d.c1, d.c2, d.c3, d.c4, p.score
  FROM PREDICT(MODEL = @model, DATA = @input as d)
  WITH(score float) as p;
```

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DEMO

MACHINE LEARNING SERVICES FOR SQL SERVER

SSIS and ADF v.2

Current State: Data Integration in Azure

Azure Data Factory (ADF)

Provides orchestration, data movement and monitoring services

Orchestration model: **time series processing**

Hybrid Data movement as a Service w/ many connectors

Programmatic authoring, visual monitoring (.NET, Powershell)

SSIS: server software for ETL

Focused on ETL to/from SQL Server
Scale up data transformation engine
Visual authoring of control and data flow
Rich ecosystem (BIML, Task Libraries, etc)

ADF v2 Public Preview

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Data Factory

Managed Data Integration Service

New Pipeline Model

Rich pipeline orchestration

Triggers – ondemand, schedule, event

Data Movement as a Service

Cloud, Hybrid

30 connectors provided

SSIS Package Execution

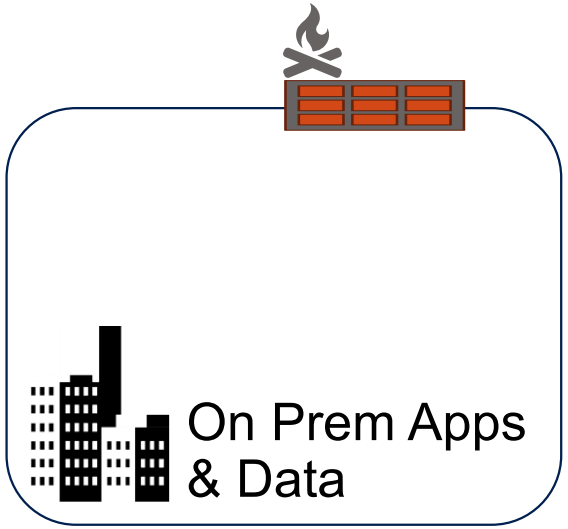
In a **managed cloud** environment

Use familiar tools, SSMS & SSDT

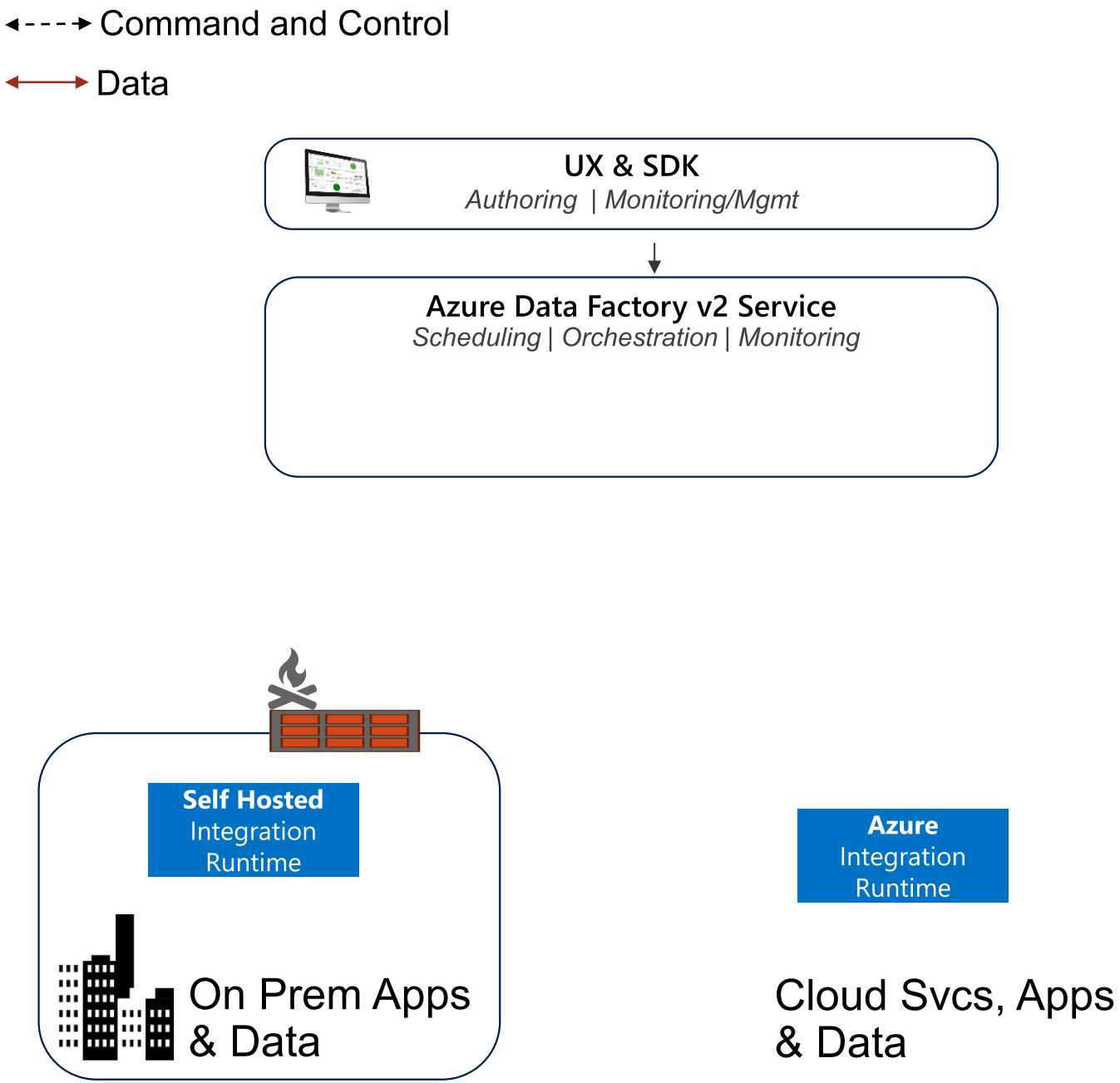
Author & Monitor

Programmability (Python, .NET, Powershell, etc)

Visual Tools (coming soon)



Cloud Svcs, Apps
& Data



Data Factory

A data integration account

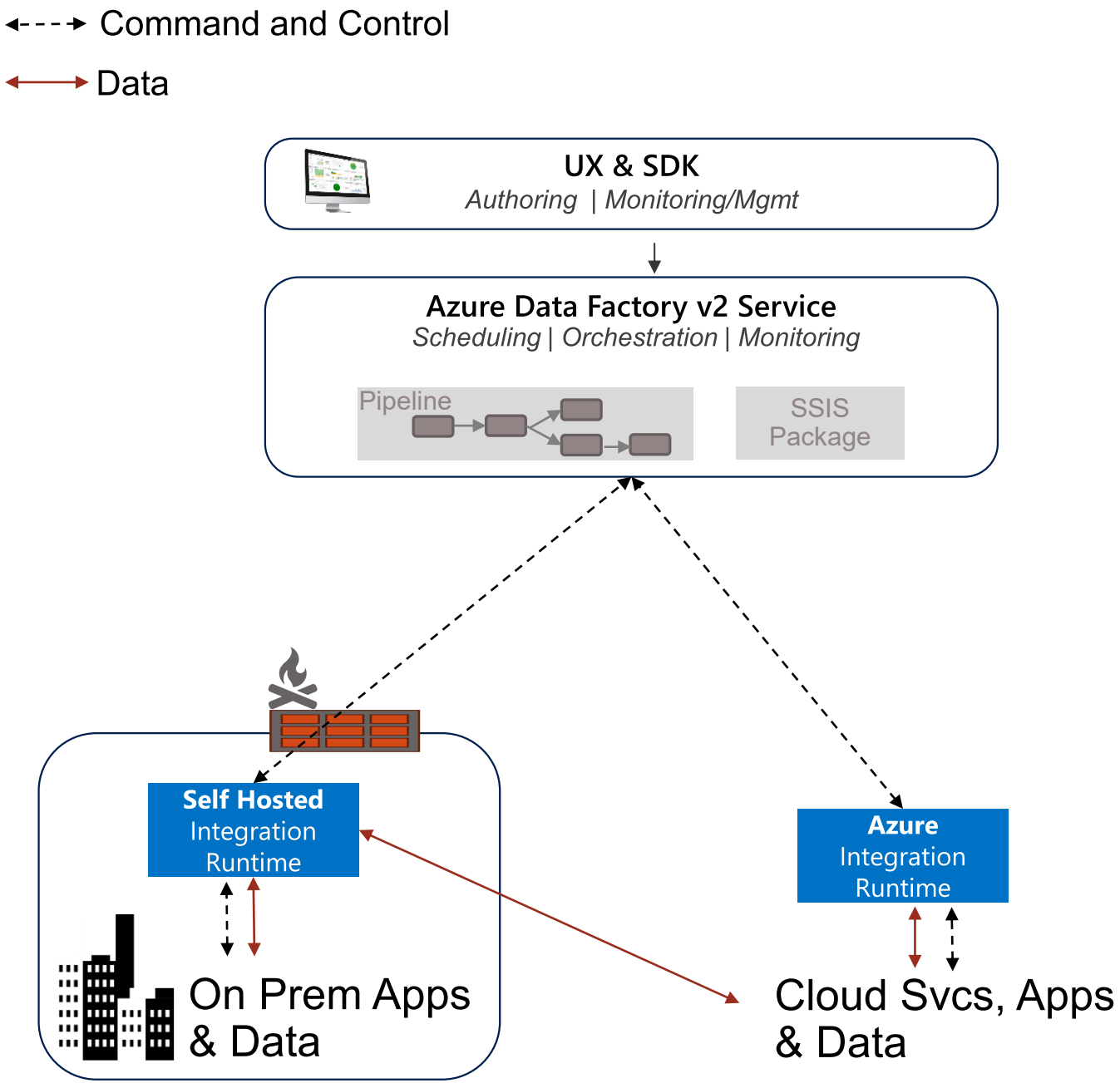
Location of orchestration, service metadata

Integration Runtime (IR)

ADF's execution engine

Three core capabilities:

- data movement
- pipeline activity execution
- SSIS package execution



Data Factory

A data integration account.

Location of orchestration, service metadata

Integration Runtime (IR)

ADF's execution engine

Three core capabilities:

- data movement
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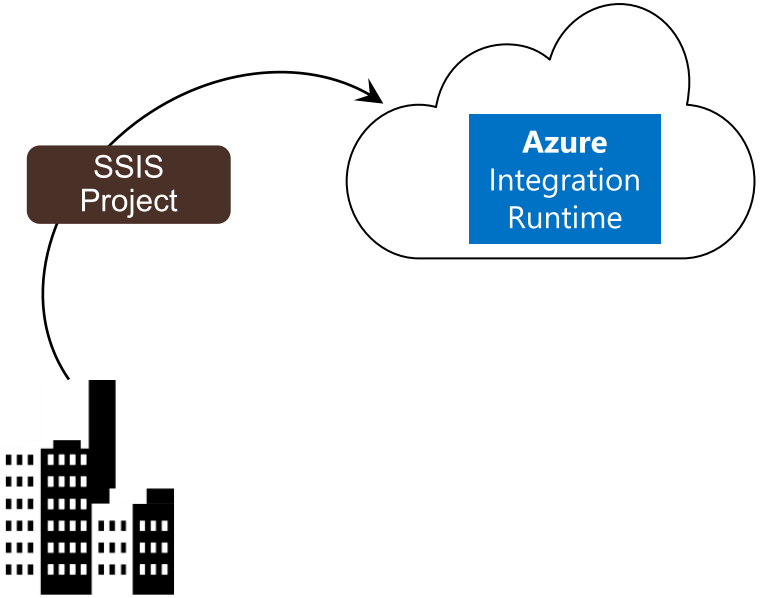
SSIS in ADFv2

LIFT EXISTING SSIS PROJECTS TO THE CLOUD

Integration Runtime for SSIS



Managed Cloud Environment

Pick # nodes & node size

Resizable

SQL Standard Edition, Enterprise coming soon

Compatible

Same SSIS runtime across Windows, Linux, Azure Cloud

SSIS + SQL Server

SQL Managed instance + SSIS (in ADFv2)

Access on premises data via VNet

Get Started

Hourly pricing (no SQL Server license required)

Use existing license (coming soon)

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DEMO

SSIS IN AZURE SQL DB AND WITH ADF INTEGRATION RUNTIME

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Database Engine

WHAT'S NEW IN SQL SERVER 2017

Installation

Separate downloads

- SQL Server Management Studio 17.x (use it also for SQL 2016)
- SQL Server Data Tools
- **New in 2017: Reporting Services**

Tempdb configuration

- Allows up to 256GB file size
- Warning if > 1GB and Instant File Initialization not enable

Linux

- Easy: import repo GPG keys, register repository, apt-get update/install
- **mssql-conf** to script out installation or to be used in a script installation
- can use environment variables

CLR Strict Security

Code Access Security (CAS) in.NET Framework

- No more supported as security boundary
- Assembly with PERMISSION_SET = SAFE may access external resources, call unmanaged code, acquire sysadmin privileges

New configuration option “CLR Strict Security”

- Treats SAFE/EXTERNAL ACCESS as UNSAFE
- Requires assemblies to be signed and login with UNSAFE ASSEMBLY permission
  - (database with TRUSTWORTHY ON owned by login with UNSAFE ASSEMBLY permission)
- Enabled by default can be disabled for backward compatibility (not recommended)

White-listing

- **sp_add_trusted_assembly**, **sp_drop_trusted_assembly**

Database Configuration

ALTER DATABASE SCOPED CONFIGURATION

- Clear procedure cache
- Cardinality estimation independent of compatibility level
- Parameter sniffing
- Query optimization hotfixes
- **New in 2017: identity cache, turn off to avoid gaps (like Trace Flag 272)**

DATABASE SCOPED CREDENTIAL

- Now a securable class supporting CONTROL, ALTER, REFERENCES, TAKE OWNERSHIP, VIEW DEFINITION

Database Development

SELECT…INTO

- Supports ON keyword to specify filegroup

New bulk access options

- FORMAT = 'CSV'
  - Support for RFC 4180 in BULK INSERT and OPENROWSET
- CREATE EXTERNAL DATA SOURCE … TYPE = BLOB_STORAGE
  - Point to specific LOCATION URL
  - Use Shared Access Signatures (SAS) via CREDENTIAL

T-SQL Development

CONCAT_WS

- Concatenate strings (columns/values) using first one as separator

TRANSLATE

- Replaces set of characters in string with other set of character
- e.g. **SELECT TRANSLATE('2*[3+4]/{7-2}', '[]{}', '()()’)**;

TRIM

- Finally!!!! ☺
- Can trim a set of characters

STRING_AGG

- Concatenates strings (rows) using separator

Temporal Tables

Support for CASCADE DELETE and CASCADE UPDATE

Retention Policy support

- Instead of custom cleanup script, stretch database or table partitioning
- Enabled at database level
  - ALTER DATABASE SET … **TEMPORAL_HISTORY_RETENTION ON**
- Specified per table
  - e.g. WITH(SYSTEM_VERSIONING = ON(… **HISTORY_RETENTION_PERIOD = 6 MONTHS**)
  - Support for DAYS, WEEKS, MONTHS, YEARS and INFINITE
  - Checked against column representing end of SYSTEM_TIME period

Columnstore Indexes in SQL Server 2016

Support for RCSI and SI isolation levels

Index defragmentation with index reorganize

Support for AlwaysOn AG readable secondary replica

Aggregate and string predicate pushdown (during scans)

Available in all lower editions with SQL Server 2016 Service Pack 1

Feature	RTM				SP1			
	Standard	Web	Express	Local DB	Standard	Web	Express	Local DB
Row-level security	Yes	No	No	No	Yes	Yes	Yes	Yes
Dynamic Data Masking	Yes	No	No	No	Yes	Yes	Yes	Yes
Change data capture*	No	No	No	No	Yes	Yes	No*	No*
Database snapshot	No	No	No	No	Yes	Yes	Yes	Yes
Columnstore	No	No	No	No	Yes	Yes	Yes	Yes
Partitioning	No	No	No	No	Yes	Yes	Yes	Yes
Compression	No	No	No	No	Yes	Yes	Yes	Yes
In Memory OLTP	No	No	No	No	Yes	Yes	Yes	No**
Always Encrypted	No	No	No	No	Yes	Yes	Yes	Yes
PolyBase	No	No	No	No	Yes	Yes	Yes	No
Fine grained auditing	No	No	No	No	Yes	Yes	Yes	Yes
Multiple filestream containers	No	No	No	No	Yes	Yes	Yes	No**

SQL Server 2016 SP1 Edition	Columnstore Index Quota (per Instance
Express	352 MB
Web	16 GB
Standard	32 GB
Developer	Unlimited
Enterprise	Unlimited

Columnstore Indexes in SQL Server 2017

Support for non-persisted computed column

Support for LOBs

- Clustered Columnstore only
- Already available also in Azure SQL Database (Premium)
- E.g. JSON data case
  - https://blogs.msdn.microsoft.com/sqlserverstorageengine/2017/02/09/json-data-in-clustered-column-store-indexes

Online non-clustered Columnstore index build/rebuild support

Feature summary for product releases

- https://docs.microsoft.com/en-us/sql/relational-databases/indexes/columnstore-indexes-what-s-new

Memory-Optimized Tables in SQL Server 2017

Support for computed columns

- Including indexes on computed columns

Support for JSON functions

- Both natively compiled T-SQL modules and check constraints

Increased support in natively compiled T-SQL modules

- CROSS APPLY, CASE, TOP (N) WITH TIES, sp_spaceused, sp_rename

Azure Storage support for memory-optimized files

- Including backup/restore

Performance

- Faster ALTER TABLE and bw-tree index rebuild during recovery
- Transaction-log REDO done in parallel

System Metadata

sys.dm_os_sys_info

- Added **socket_count, cores_per_socket, numa_node_count**
- Useful with VM

sys.dm_os_host_info

- OS information on Windows/Linux

Undocumented

- **sys.dm_os_enumerate_fixed_drives**
- **sys.dm_os_enumerate_filesystem(dir, pattern)**
- **sys.dm_os_file_exists(dir_or_file)**

Database Metadata

sys.dm_db_log_stats()

- Summary info about t-log, useful for monitoring

sys.dm_db_file_space_usage

- Added **modified_extent_page_count**
- Useful to build smart backup solutions eg. differential if < 70-80%, full otherwise

sys.dm_tran_version_store_space_usage

- Reserved space in pages/KB

sys.dm_db_stats_histogram(object, stat)

- SQL Server 2017 / SQL Server 2016 SP1 CU2

DBCC CLONEDATABASE

Creates empty copy of database for troubleshooting

- No data but full schema (constraints, type, t-sql modules, etc.)
- Statistics
- Non blocking
- Read only by default but can be changed
- Optionally NO_STATISTICS, NO_QUERYSTORE
- SQL Server 2012 SP4, 2014 SP2 CU3, 2016 SP1, 2017

USE HINT Query Option

Named hints instead of hard to remember trace flags

- **OPTION (USE HINT('DISABLE_PARAMETER_SNIFFING'))** instead of **OPTION (QUERYTRACEON 4136)**
- Doesn't require sysadmin permission!
- Available in SQL Server 2017 / SQL Server 2016 SP1

List from **sys.dm_exec_valid_use_hints**

ASSUME_JOIN_PREDICATE_DEPENDS_ON_FILTERS	9476
ASSUME_MIN_SELECTIVITY_FOR_FILTER_ESTIMATES	4137, 9471
DISABLE_PARAMETER_SNIFFING	4136
DISABLE_OPTIMIZER_ROWGOAL	4138
DISABLE_OPTIMIZED_NESTED_LOOP	2340
ENABLE_HIST_AMENDMENT_FOR_ASC_KEYS	2389
ENABLE_QUERY_OPTIMIZER_HOTFIXES	4199
FORCE_DEFAULT_CARDINALITY_ESTIMATION	2312
FORCE_LEGACY_CARDINALITY_ESTIMATION	9481

Execution Plan Enhancements

Shows query time statistics

- CPU time and Elapsed time
- SQL Server 2012 SP4, 2016 SP1, 2017, 2014 (TBD)

Shows query-level Wait Stats!!

- Top 10 wait stats
- Also tracked in **sys.dm_exec_session_wait_stats**
- SQL Server 2012 SP4, 2016 SP1, 2017 (compatibility 140!), 2014 (TBD)

Shows query-Level Trace Flags

- Including level: global, session or query
- SQL Server 2012 SP4, 2014 SP2, 2016 SP1, 2017

Query execution profiling

sys.dm_exec_query_statistics_xml(session)

- Execution plan with transient, in-flight statistics (e.g. row count, CPU)
- Available in SQL Server 2017 / SQL Server 2016 SP1
- Works both with standard and lightweight statistics profiling
  - E.g. execute SET STATISTICS XML ON before query to analyze

Query Store in SQL Server 2017

Wait Statistics integrated in Query Store!

- sys.query_store_wait_stats
- Organized in Wait Categories
- ALTER DATABASE …
  SET QUERY_STORE (WAIT_STATS_CAPTURE_MODE = ON)

Automatic Tuning

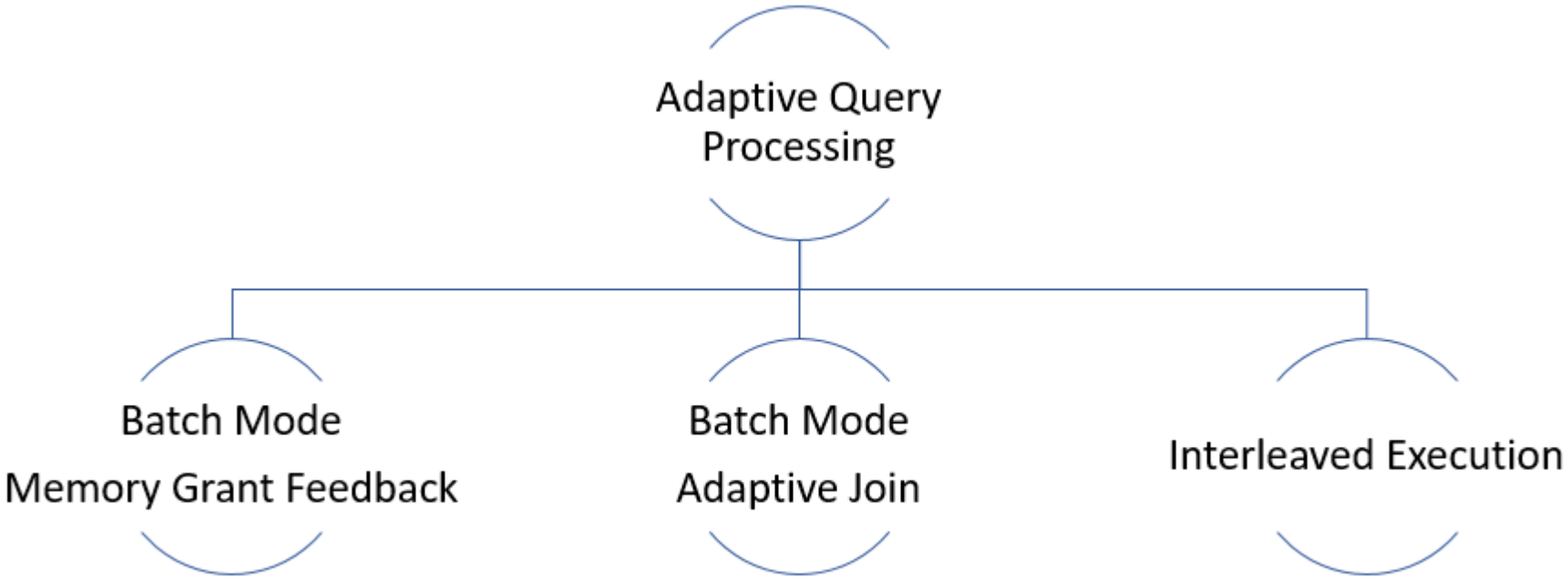
Automatic plan correction

- Automatically force last good plan when regression detected
- Minimum 10 CPU seconds improvement
- sys.dm_db_tuning_recommendations for manual corrections
- SQL Server 2017 and Azure SQL Database

Automatic index management

- Automatically CREATE and DROP indexes
- Verifies usage and performance improvements
- Azure SQL Database only

Adaptive Query Processing

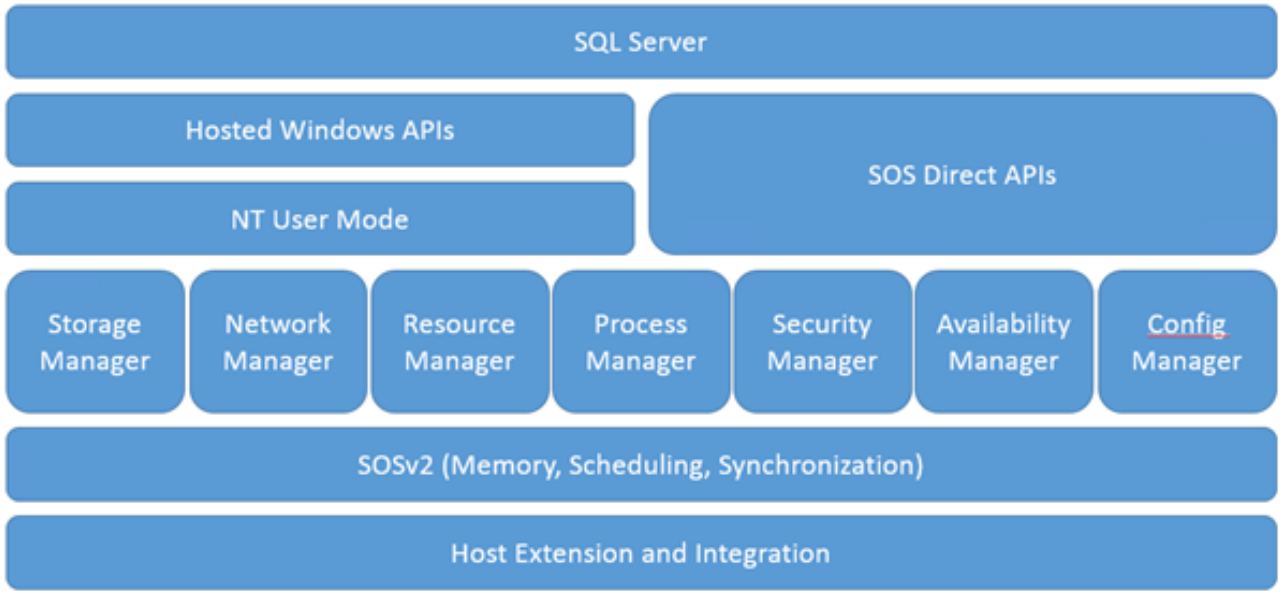


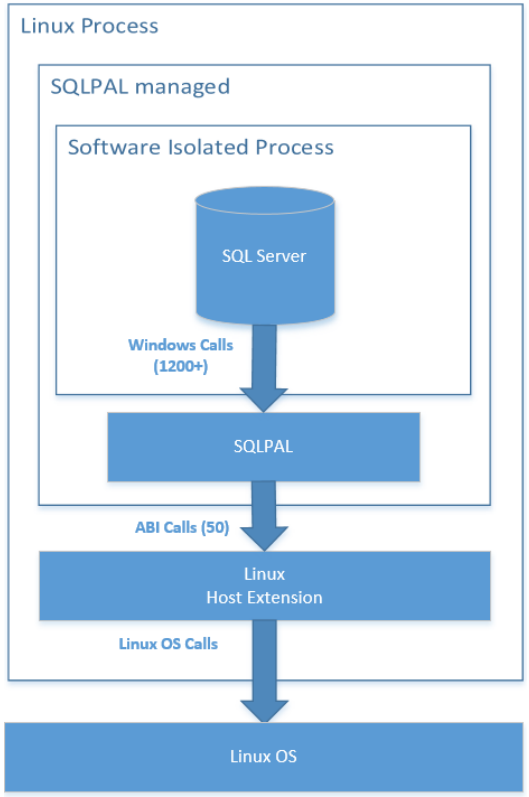
OverNet EDUCATION

SQL on Linux

WHAT'S NEW IN SQL SERVER 2017

System Architecture





Supported versions

Red Hat Enterprise Linux 7.3 or 7.4

Ubuntu Linux 16.04 LTS

Suse Linux Enterprise Server v12 SP2

Docker Engine 1.8+ on Linux, Mac, Windows

Minimum requirements

2 x core CPU 2Ghz (x64 only)

3,25 GB memory

6GB disk space (XFS or EXT4 file systems)

Packages

mssql-tools

- ODBC libraries
- sqlcmd command-line querying tool
- bcp for bulk import/export

mssql-server-agent

mssql-server-fts

- Full-Text indexing and Semantic Search (requires additional steps)

mssql-server-is

Tools

In Linux

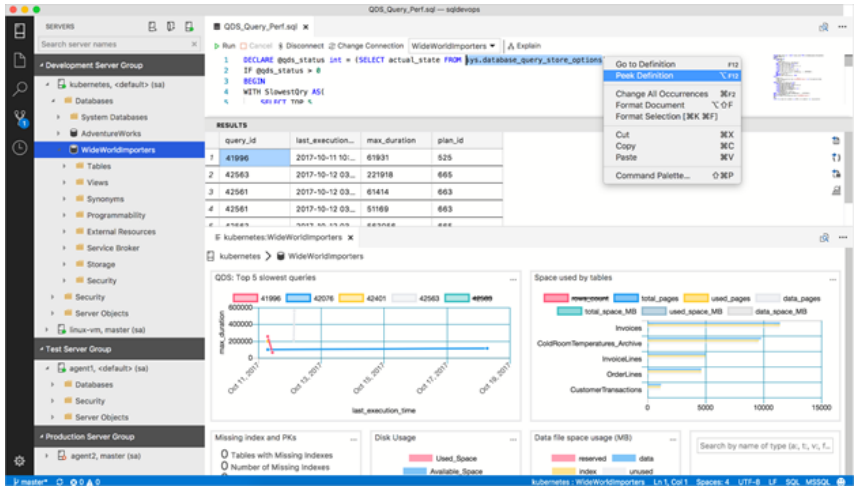
- Configuration script **mssql-conf**
- T-SQL commands with **sqlcmd**

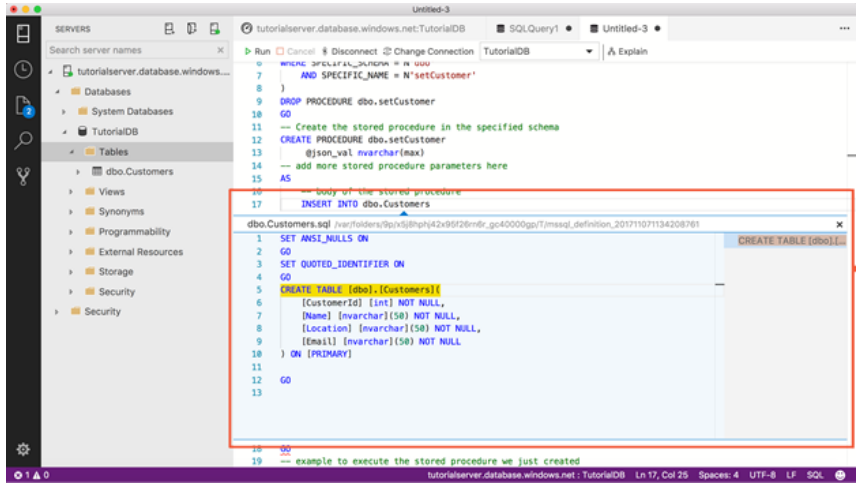
In Windows

- SQL Server Management Studio (SSMS)
- Visual Studio with SQL Server Data Tools (SSDT)
- Visual Studio Code with **mssql** extension
- PowerShell
- T-SQL commands with **sqlcmd**

Multi platform (Windows, Mac, Linux)

- SQL Operation Studio (preview)





Sessione di approfondimento

Data, Analytics & IoT - DEV015 - SQL Server su Linux: dalla L alla X

Danilo Dominici

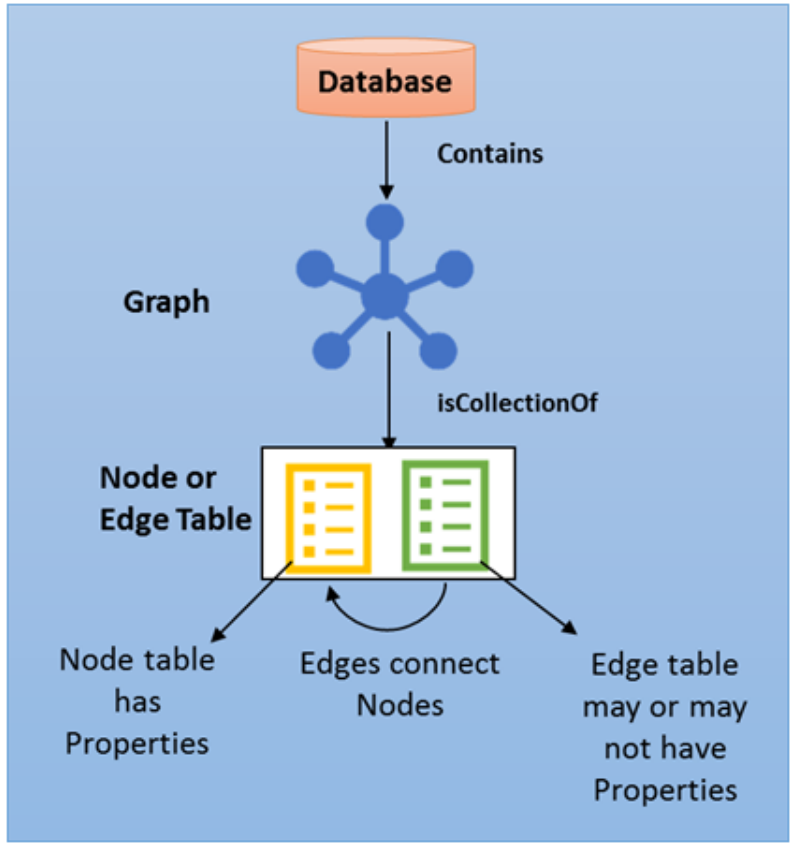
Livello: 200
Giorno: 29 Novembre - Orario: 15:15-16:30 - Sala: Gialla
Track: Data, Analytics & IoT

OverNet EDUCATION

Graph Processing

WHAT'S NEW IN SQL SERVER 2017

Architecture



Node Table

- Represent an entity

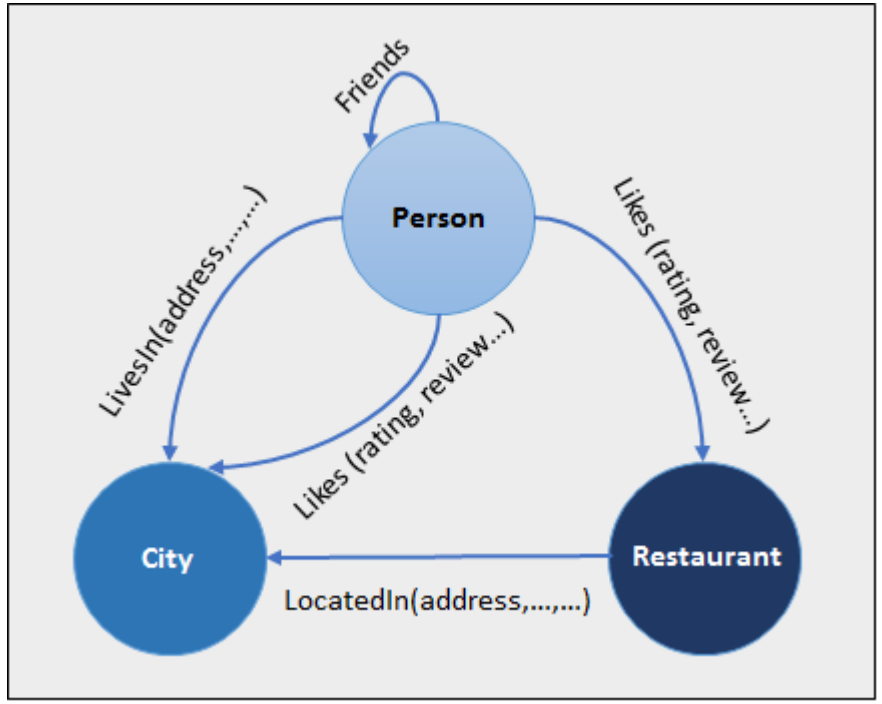
Edge Table

- Represent many-to-many relationship
- May have properties
- Is directed, connects two nodes

MATCH T-SQL Command

- Search condition for graph objects
- Pattern matching and traversal

Sample


Friends
Person
LivesIn(address,...,...)
Likes (rating, review...)
Likes (rating, review...)
City
Restaurant
LocatedIn(address,...,...)

Sessione di approfondimento

DEV030 - [Graph DB] in SQL Server 2017, Azure SQLDB e Azure CosmosDB : da 0 a 100 in 60 minuti

Igor Pagliai

Livello: 300
Giorno: 29 Novembre - Orario: 12:15-13:30 - Sala: Verde
Track: Data, Analytics & IoT

SQL Server HA and DR solutions

Simple

VM failover
- Resilience against guest and OS-level failures
- Minutes RTO

Backup and restore
- DR
- Minutes to hours RTO

Standard

AlwaysOn Failover cluster
- Instance-level protection
- Seconds to minutes RTO

AlwaysOn Basic availability groups
- AG with 2 replicas

Log shipping
- Warm standbys for DR

Mission critical

AlwaysOn Availability Groups
- Database level protection
- Seconds to minutes RTO

AlwaysOn Distributed Availability Groups
- DR
- Seconds to minutes RTO

AlwaysOn FCI in newer releases

SQL Server 2016

- Group Managed Service Accounts (gMSA)
  - Managed directly by AD
  - Automatic password rotation

SQL Server 2017

- SQL Server on Linux support

AlwaysOn AG in SQL Server 2017

Cross database transactions supported (MSDTC)

Minimum number of (sync) replicas to commit

Clusterless support

- CLUSTER_TYPE = NONE (e.g. read-only scale out replicas not used for HA)

SQL Server on Linux support

- CLUSTER_TYPE = EXTERNAL (Pacemaker)
- CLUSTER_TYPE = NONE
- Allows Windows-Linux cross-OS migrations

WPC2017 28-29-30/NOVEMBRE MILANO

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Q&A

Domande e risposte

WPC2017 28-29-30/NOVEMBRE MILANO

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